MA341
Applied ODE's

Lecture Notes

5 Second Order ODE

A general form of second oOrder ODE is given by

$$F(x, y, y', y') = 0 \quad \text{or} \quad y'' = f(x, y, y')$$

The number of initial conditions required for uniqueness of second order ODE is two:

$$y(x_0) = y_0, y'(x_0) = y'_0$$

Example 5.1. Solve

$$y'' = (y')^2.$$

Denote $v = y'$, substitute into equation to get

$$\begin{aligned} v' &= v^2 \\ \frac{dv}{v^2} &= dx \\ -\frac{1}{v} &= x + C_1 \\ v &= -\frac{1}{x + C_1} \end{aligned}$$

Thus

$$y' = v = -\frac{1}{x + C_1},$$

and we integrate again to get

$$y = -\int \frac{1}{x + C_1} = -\ln x + C_1 + C_2.$$

5.1 Linear Second Order ODE

$$y'' + p(x)\,y' + q(x)\,y = g(x)$$

Example 5.2. Some famous second order ODEs:

1. Legender's equation:

$$\left(1 - x^2\right) y'' - 2xy' + \alpha\,(\alpha + 1)\,y = 0$$

2. Bessel equation:

$$x^2 y'' + xy' + \left(x^2 - v^2\right) y = 0$$

Theorem 5.3. Suppose $p(x)$ $q(x)$, $g(x)$ are continuous function on (a, b). Let $x_0 \in (a, b)$, then there is a unique solution on (a, b) that satisfies the equation

$$y'' + p(x)\,y' + q(x)\,y = g(x)\,,$$

and the initial conditions

$$y(x_0) = y_0,\ y'(x_0) = y_0',$$

where y_0, y_0' are given constants.

5.1.1 Homogeneous second order linear equation

Homogeneous equation has $g = 0$, so consider

$$y'' + p(x)\,y' + q(x)\,y = 0$$

Definition 5.4. Given functions $f_1(x), f_2(x), \ldots, f_k(x)$ defined in $I = [a, b]$. A sum of the form

$$\sum_{i=1}^{k} c_i f_i(x) = c_1 f_1(x) + c_2 f_2(x) + \cdots + c_3 f_k(x), \quad c_i \in \mathbb{R}$$

is called a **linear combination** of $f_1(x), f_2(x), \ldots, f_k(x)$.

Example 5.5. Consider e^x, $\sin x$, $\ln x$. Example of linear combination:

$$3e^x + 6\sin x + \ln x.$$

Theorem 5.6. Superposition principle. Given $y_1(x)$, $y_2(x)$ solutions to the second order linear *homogeneous* equation, then a linear combination of solutions $y_1(x)$, $y_2(x)$:

$$y(x) = c_1 y_1(x) + c_2 y_2(x)$$

is also solution to the equation.
Proof:

$$\begin{aligned} y(x) &= c_1 y_1(x) + c_2 y_2(x) \\ y'(x) &= c_1 y_1'(x) + c_2 y_2'(x) \\ y''(x) &= c_1 {y''}_1(x) + c_2 {y''}_2(x). \end{aligned}$$

Hence

$$\begin{aligned} y'' + p(x)\, y' + q(x)\, y &= \underbrace{\left(c_1 {y''}_1 + c_2 {y''}_2\right)}_{y''} + p(x) \underbrace{\left(c_1 y_1' + c_2 y_2'\right)}_{y'} + q(x) \underbrace{\left(c_1 y_1 + c_2 y_2\right)}_{y} \\ &= c_1 \underbrace{\left({y''}_1 + p(x)\, y_1' + q(x)\, y_1\right)}_{=0} + c_2 \underbrace{\left({y''}_2 + p(x)\, y_2' + q(x)\, y_2\right)}_{=0} = 0 \end{aligned}$$

Example 5.7. Consider $y'' - y = 0$.

1. Show that $y_1 = e^x$ and $_2 = e^{-x}$ solves the equation.
2. Show that $\cosh x = \frac{e^x + e^{-x}}{2}$ also solves the equation.

Solution: Substitute all the options into equation to see.

Example 5.8. Consider $y'' - y = x$.

1. Show that $y = x$solves the equation.
2. Show that $y = -2x$ doesn't solve the equation.
3. Explain why this is not a contradiction to the superposition principle (5.6).

Solution: Substitute all the options into equation to see. This example doesn't contradict the superposition principle since the equation in this example is not homogeneous.

Theorem 5.9. Assume $p(x)$, $q(x)$ are continuous functions in an open interval I, and $y_1(x)$, $y_2(x)$ are solutions of

$$y'' + p(x)\,y' + q(x)\,y = 0.$$

If

$$y_1(x_0)\,y_2'(x_0) - y_1'(x_0)\,y_2(x_0) \neq 0,$$

then

$$y(x) = c_1 y_1(x) + c_2 y_2(x)$$

is a general solution of the equation.

Definition 5.10. Wronskian:

$$\begin{aligned} W(y_1, y_2, x) &= [W(y_1, y_2)](x) = W(y_1, y_2)(x) = W_{1,2}(x) \\ &= \det \begin{pmatrix} y_1(x) & y_2(x) \\ y_1'(x) & y_2'(x) \end{pmatrix} \\ &= y_1(x)\,y_2'(x) - y_2(x)\,y_1'(x) \end{aligned}$$

Similarly,

$$W(f_1, \ldots, f_k, x) = \det \begin{pmatrix} f_1(x) & \cdots & f_k(x) \\ f_1'(x) & \cdots & f_k'(x) \\ \vdots & & \\ f_1^{(k)}(x) & \cdots & f_k^{(k)}(x) \end{pmatrix}.$$

Theorem 5.11. Let $p(x)$, $q(x)$ be continuous functions in an open interval I, and $y_1(x)$, $y_2(x)$ are solutions of

$$y'' + p(x)\,y' + q(x)\,y = 0.$$

If

$$W(y_1, y_2, x) \neq 0,$$

then

$$y(x) = c_1 y_1(x) + c_2 y_2(x)$$

is a general solution of the equation.

Example 5.12. Consider $y'' + y = 0$. Show that $y_1 = \sin x$ and $y_2 = \cos x$ solves the equation. Next, show that the Wronskian doesn't vanish and write a general solution.

$$y_1' = \cos x,\ \ {y''}_1 = -\sin x \Rightarrow {y''}_1 + y_1 = -\sin x + \sin x = 0.$$

Similarly,

$$y_2' = -\sin x,\ \ {y''}_2 = -\cos x \Rightarrow {y''}_2 + y_2 = -\cos x + \cos x = 0.$$

Thus,

$$W(y_1, y_2, x) = y_1 y_2' - y_1' y_2 = -\sin x \sin x - \cos x \cos x = -\left(\cos^2 x + \sin^2 x\right) = -1 \neq 0,$$

hence the general solution is

$$y(x) = c_1 \cos x + c_2 \sin x.$$

Definition 5.13. The solutions

$$y_1(x), y_2(x)$$

that satisfies

$$W(y_1, y_2, x) \neq 0$$

called a **fundamental set** - a basis of the solution and any other solution can be created by linear combination of them.

Theorem 5.14. Abel's Formula. Let $y_1(x)$, $y_2(x)$ be solutions to

$$y'' + p(x) y' + q(x) y = 0,$$

where $p(x)$, $q(x)$ be continuous functions in an open interval I, then

$$W_{1,2}(x) = Ce^{-\int p(t)dt}$$

Furthermore,

- The constant C is independent of x, but depends on

$$y_1(x), y_2(x).$$

- Wronskian is vanishing if and only if $C = 0$.

Example 5.15. Consider

$$2x^2y'' + 3xy' - y = 0.$$

Rewrite it as

$$y'' + \underbrace{\frac{3x}{2x^2}}_{p} y' - \frac{1}{2x^2}y = 0.$$

- Show that $y_1 = x^{1/2}, y_2 = x^{-1}$ solve the equation.
- Calculate Wronskian using determinant formula:

$$\begin{aligned} W_{1,2} &= y_1'y_2 - y_1y_2' \\ &= \underbrace{x^{1/2}}_{y_1}\underbrace{\left(-x^{-2}\right)}_{y_2'} - \underbrace{\frac{1}{2}x^{-1/2}}_{y'_1}\underbrace{x^{-1}}_{y_2} \\ &= -\left(x^{-1.5} + \frac{1}{2}x^{-1.5}\right) \\ &= -1.5x^{-1.5} \end{aligned}$$

- Calculate Wronskian using Abel's formula

$$
\begin{aligned}
W_{1,2}(x) &= ce^{-\int^x p(t)dt} \\
&= ce^{-\frac{3}{2}\int \frac{1}{t}dt} \\
&= ce^{-\frac{3}{2}\ln x} = ce^{\ln x^{-3/2}} \\
&= cx^{-1.5}
\end{aligned}
$$

for the given solution $c = -1.5$.

- Assume now that only $y_1 = x^{1/2}$ is known, find y_2

$$
\begin{aligned}
cx^{-1.5} = W_{1,2} &= y_1' y_2 - y_1 y_2' \\
&= \underbrace{\frac{1}{2}x^{-1/2}}_{y'_1} y_2 - \underbrace{x^{1/2}}_{y_1} y_2'
\end{aligned}
$$

Solve it for y_2 to get a first order ODE:

$$
\begin{aligned}
x^{1/2}y_2' &= \frac{1}{2}x^{-1/2}y_2 - cx^{-1.5} \\
y_2' - \frac{1}{2}x^{-1}y_2 &= -cx^{-2},
\end{aligned}
$$

which solution is of course $y_2 =^{-1}$ (verify!).

Definition 5.16. The functions

$$
f_1(x), f_2(x), ..., f_n(x)
$$

are **linearly dependent** if there is a vanishing linear combination

$$
c_1 f_1(x) + c_2 f_2(x) + ... + c_n f_n(x) = 0
$$

where not all $c_i = 0$. Otherwise, i. e. in the case the linear combination above is vanishing only when all $c_i = 0$, they are **linearly independent dependent**.

Intuition: If we can express one of the functions as linear combination of the others

$$f_j\left(x\right) = c_1 f_1\left(x\right) + ... + c_{j-1} f_2\left(x\right) + c_{j+1} f_2\left(x\right) + ... + c_n f_n\left(x\right)$$

then this set is linearly dependent.

Example 5.17. • $\{2x, 4x\}$ is linearly dependent.

- e^x, e^{2x} is linearly independent.

Theorem 5.18. If

$$f_1\left(x\right), f_2\left(x\right), ..., f_n\left(x\right)$$

are linearly dependent then

$$W_{1,2...,n}\left(x\right) = 0,\ \forall x \in I,$$

but,

$$W_{1,2...,n}\left(x\right) = 0,\ \forall x \in I$$

it is *insufficient condition* to conclude that

$$f_1\left(x\right), f_2\left(x\right), ..., f_n\left(x\right)$$

are linearly dependent. However, if $y_1(x)$, $y_2(x)$ are solutions to second order linear homogeneous equation, and

$$W_{1,2...,n}\left(x\right) \neq 0,\ \forall x \in I$$

then $y_1(x)$, $y_2(x)$ are linearly independent.

Example 5.19. Consider the following functions

$$f_1(x) = \begin{cases} 1 + x^3 & x \leq 0 \\ 1 & x > 9 \end{cases}$$

$$f_2(x) = \begin{cases} 1 & x \leq 0 \\ 1 + x^3 & x > 9 \end{cases}$$

$$f_3(x) = 3 + x^3$$

- The functions f_1,f_2,f_3 are linearly independent in $(-1, 1)$:

  If $-1 < x \leq 0$, assuming there are c_1, c_2, c_3 that are not all zeros and satisfies

$$c_1 f_1(x) + c_2 f_2(x) + c_3 f_3(x) = c_1(1 + x^3) + c_2 + c_3(3 + x^3) = 0$$

  gives

$$\begin{aligned} c_1 + c_3 &= 0 \\ c_1 + c_2 + 3c_3 &= 0. \end{aligned}$$

  If $0 < x < 1$, the same coefficients c_1, c_2, c_3 should also satisfy

$$c_1 f_1(x) + c_2 f_2(x) + c_3 f_3(x) = c_1 + c_2(1 + x^3) + c_3(3 + x^3) = 0,$$

  which gives

$$\begin{aligned} c_2 + c_3 &= 0 \\ c_1 + c_2 + 3c_3 &= 0. \end{aligned}$$

  All together we got

$$\begin{aligned} c_1 + c_3 &= 0 \\ c_2 + c_3 &= 0 \\ c_1 + c_2 + 3c_3 &= 0. \end{aligned}$$

  The only solution to this system is $c_1 = c_2 = c_3 = 0$, hence the functions are linearly independent.

- However, $W(f_1, f_2, f_3)(x) = 0$:

$$W(f_1, f_2, f_3)(x) \underset{x \leq 0}{=} \begin{vmatrix} 1 + x^3 & 1 & 3 + x^3 \\ 3x^2 & 0 & 3x^2 \\ 6x & 0 & 6x \end{vmatrix} = -\begin{vmatrix} 3x^2 & 3x^2 \\ 6x & 6x \end{vmatrix} = 0$$

Similarly,

$$W(f_1, f_2, f_3)(x) \underset{x>0}{=} \begin{vmatrix} 1 & 1+x^3 & 3+x^3 \\ 0 & 3x^2 & 3x^2 \\ 0 & 6x & 6x \end{vmatrix} = \begin{vmatrix} 3x^2 & 3x^2 \\ 6x & 6x \end{vmatrix} = 0.$$

Hence,

$$W(f_1, f_2, f_3)(x) = 0, \forall x \in \mathbb{R}.$$

5.1.2 D'Alembert's Order Reduction Method

Consider

$$y''' + 2y'' - 3y' + y = 0.$$

Assume y_1 is a know solution to this equation. Consider solution in the form

$$y = v\left(x\right) y_1\left(x\right).$$

Differentiate it twice to get

$$\begin{aligned} y' &= v'y_1 + vy_1' \\ y'' &= v''y_1 + v'y_1' + v'y_1' + vy''_1 = v''y_1 + 2v'y_1' + vy''_1 \\ y''' &= v'''y_1 + v''y_1' + 2v''y_1' + 2v'y''_1 + v'y''_1 + vy_1''' \\ &= v'''y_1 + 3v''y_1' + 3v'y''_1 + vy_1^{(3)} \end{aligned}$$

Substitute into equation:

$$\begin{aligned} 0 &= y''' + 2y'' - 3y' + y \\ &= \left(v'''y_1 + 3v''y_1' + 3v'y''_1 + vy_1'''\right) + 2\left(v''y_1 + 2v'y_1' + vy''_1\right) - 3\left(v'y_1 + vy_1'\right) + v\left(x\right) y_1\left(x\right) \\ &= v'''y_1 + \left(3y_1' + 2y_1\right) v'' + \left(4y_1' - 3y_1 + 3y''_1\right) v' + \underbrace{\left(y_1''' + 2y''_1 - 3y_1' + y_1\right)}_{=0} v \end{aligned}$$

Substitute now $u = v'$ to get a second order ODE

$$0 = u''y_1 + (3y_1' + 2y_1)\, u' + (4y_1' - 3y_1 + 3y''_1)\, u$$

or

$$u'' = -\frac{3y_1' + 2y_1}{y_1} u' - \frac{4y_1' - 3y_1 + 3y''_1}{y_1} u$$